

ALES – Affordable Laparoscopic End-to-End Autonomous Suturing

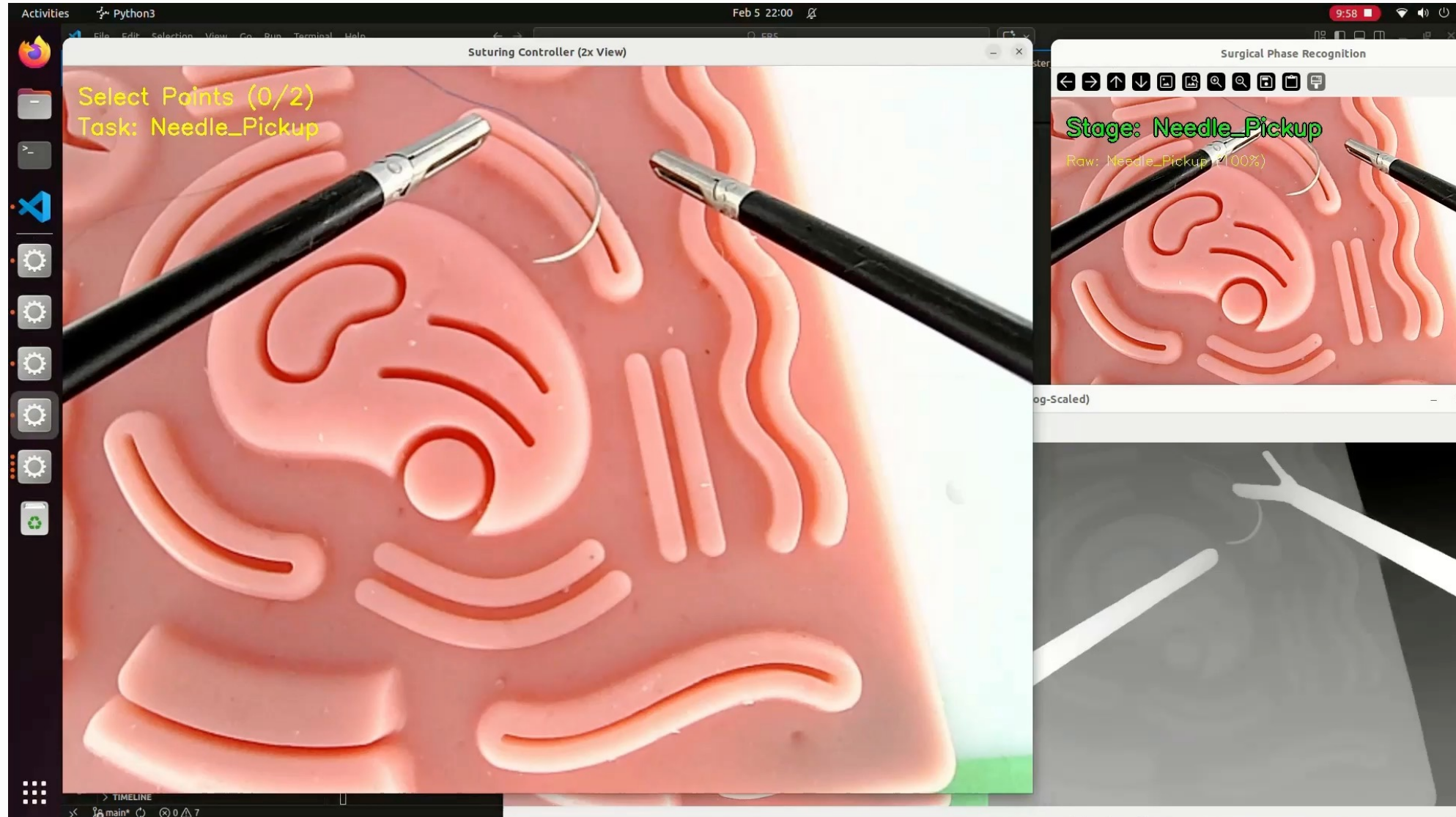
26.02.06 Dongho Yee

Contents

- **Depth Map**
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- 
- A decorative background graphic consisting of three concentric circles. A horizontal line passes through the center of the circles, with small open circles at its intersections with the inner, middle, and outer circles. The area between the middle and outer circles is shaded with vertical gray stripes.

Feb.5 VLA Development

Achieved using **only 59 episodes** for training.



Feb.5 VLA Developement

More than 100 trials – gained lots of know-hows in training VLAs

☐  NAME 72 visualized	STATE	NOTES	USER	TAGS	CREATED ▾	RUNTIME	SWEEP	_TARGET_	CHECKPOINT.SA	CHECKPOINT.SA	CHECKPOINT
☐   2026.01....phantom	 Killed	Add notes...	donghoyee	 	1w ago	7m 12s	-	diffusion_policy	true	false	epoch={ep
☐   2026.01....phantom	 Killed	Add notes...	donghoyee	 	1w ago	1d 1h 25m 35s	-	diffusion_policy	true	false	epoch={ep
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☐   2026.01....phantom	 Failed	Add notes...	donghoyee	 	1w ago	4m 5s	-	diffusion_policy	true	false	1-20 ▾ of

Data is everything - Scale, frequency, length, quality - all matters

Depth Map Refinement

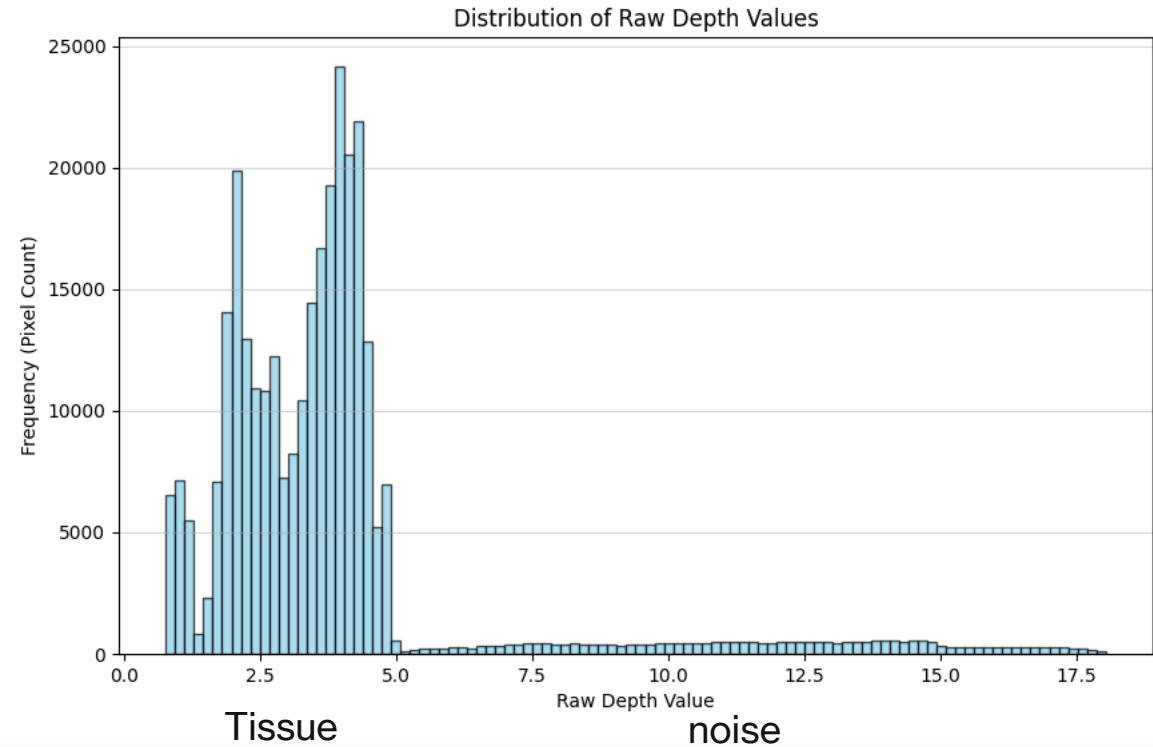
Lots of comments on Depth map. Analyzed relative depth map in detail

Previous – Used simple Min-Max normalization.

Pixel-wise histogram shows lots of noisy depth



Result of Simple Min-Max normalization
visualized using Greyscale

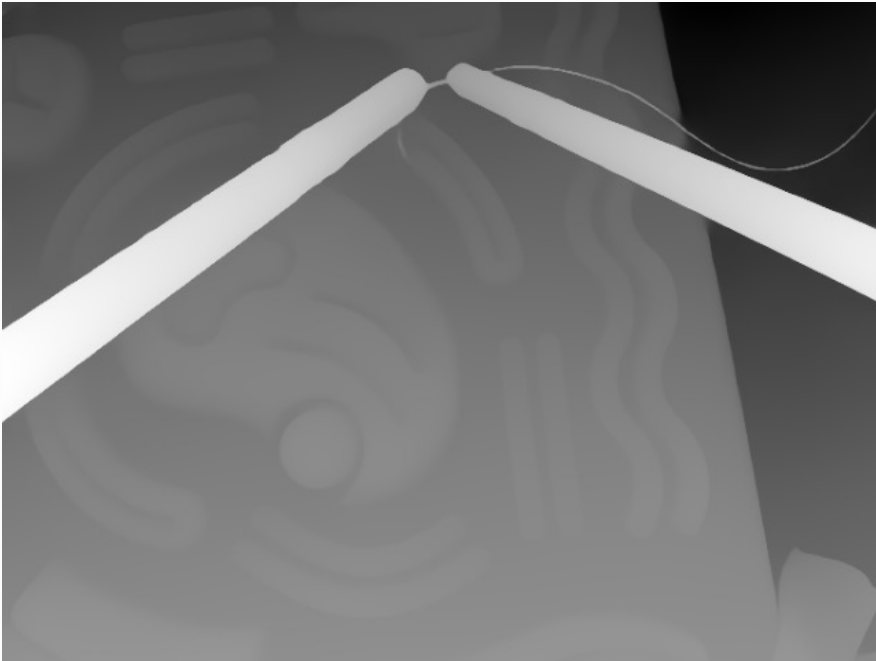


Depth Map Refinement

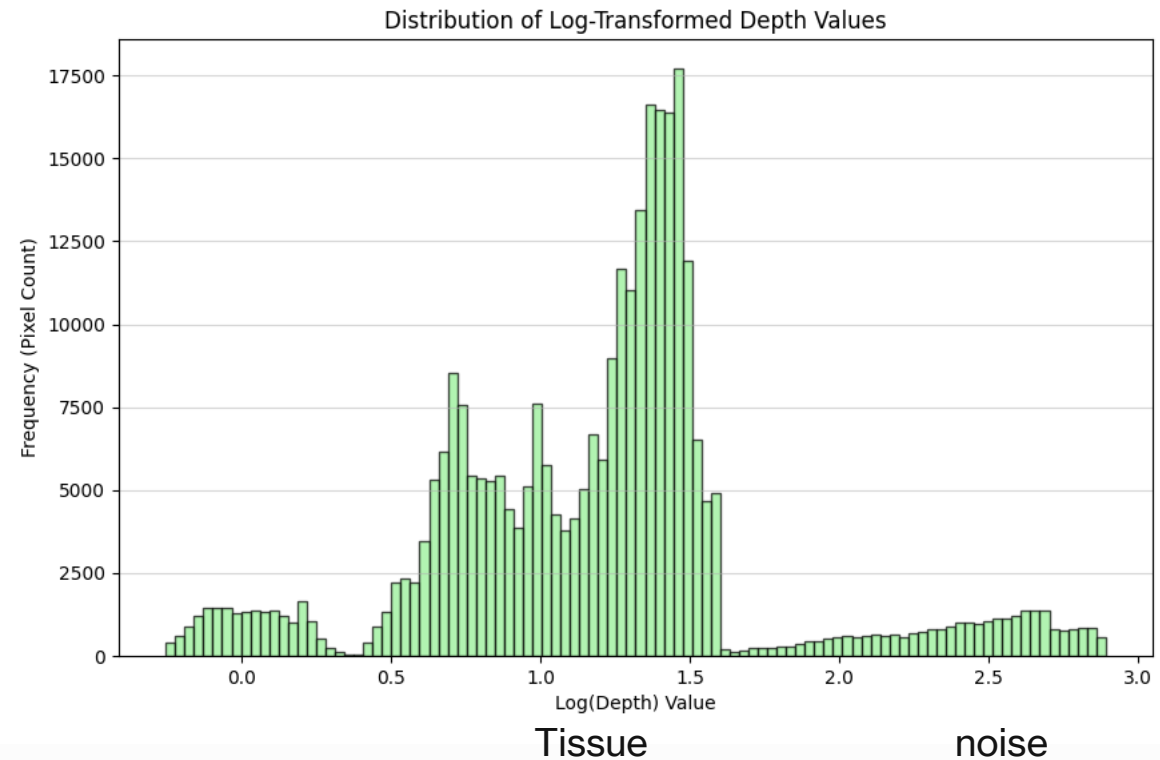
Min-Max normalization after Logscale conversion is much better

To focus more on the important tissue-part, needs to draw out noisy large depth values.

Experimented with percentile clipping, logscale conversion – Logscale conversion is more natural.

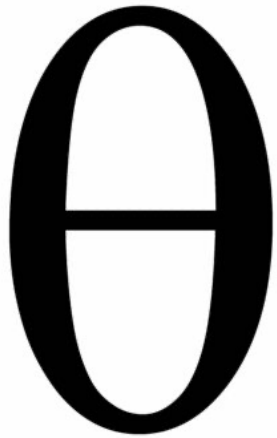


Result of Min-Max normalization after logscale conversion visualized using Greyscale

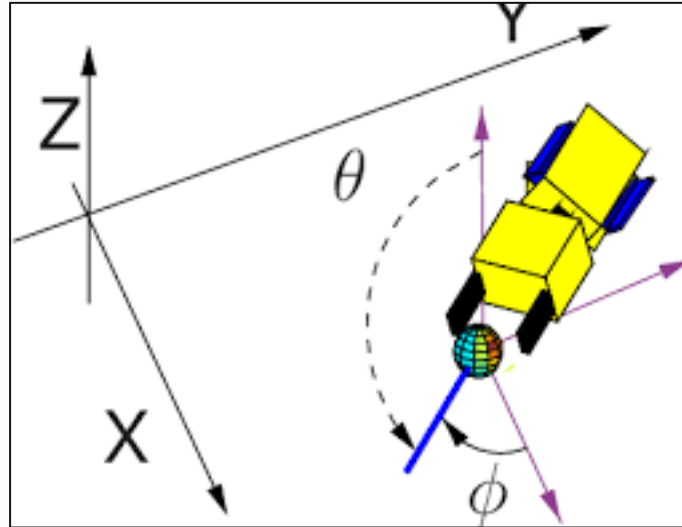


Action Types

There are mainly 3 types of actions that can be used as the goal for VLA training

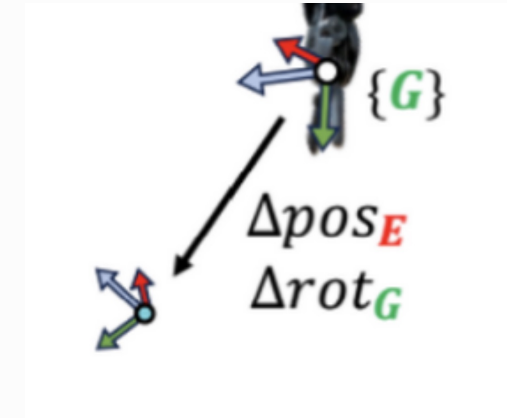


Joint Values



End-Effector

Absolute position



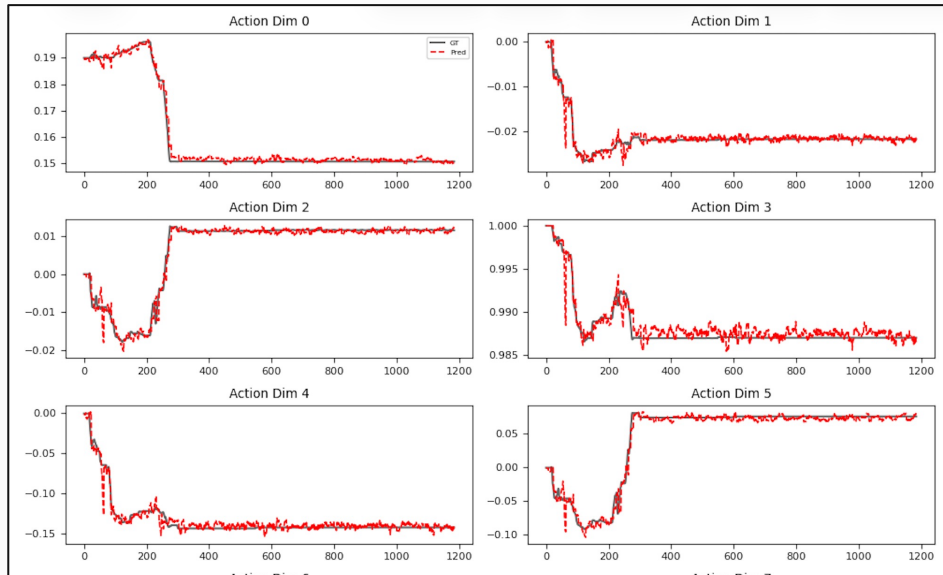
End-Effector

Delta position

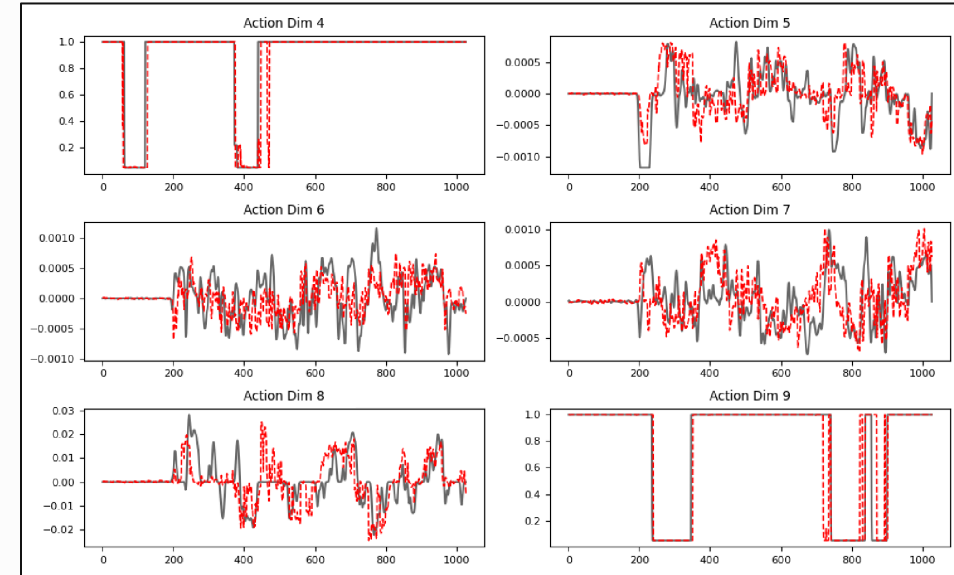
Mainly Experimented with End-Effector Absolute vs Delta position

Action Types

End-Effector **Absolute** Position Turned out to be better in training
(validation graph) but **worse** in Real-life rollouts



Absolute position validation graph



Delta position validation graph

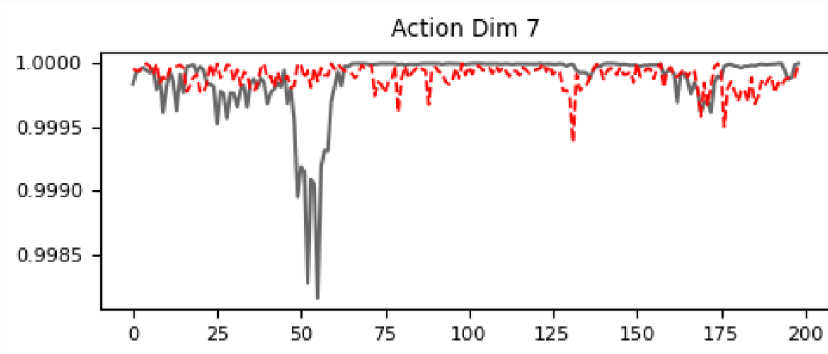
The Trap of Identity Mapping – Robot learns to predict the current state instead of predicting future state.

Action Types

How to represent difference in position – use fewer variables as possible & check for normalization (important!)

Index	Name	Format	Description
0 ~ 2	Left Delta Position	[dx, dy, dz]	현재 왼쪽 그리퍼 좌표계(Local) 기준 위치 변화량 (m) (Global x, y, z 아님)
3 ~ 8	Left Delta Rotation	[r1, r2, r3, r4, r5, r6]	현재 왼쪽 그리퍼 자세 기준 상대 회전 (Relative Rotation Matrix → 6D)

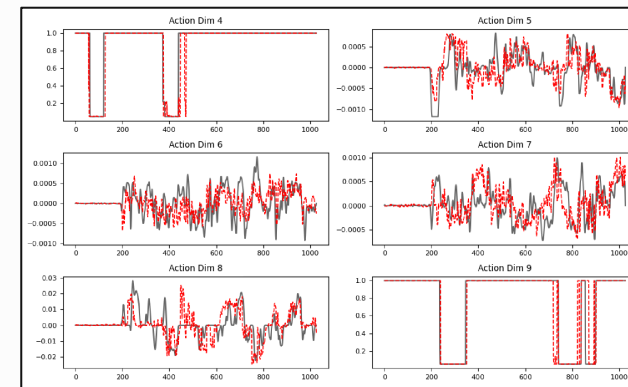
Using 9 variables for position representation



Almost stationary variable – bad for training

0	Left Δx	Left EE Local Frame 기준 X축 변화량	$P_{next} - P_{curr}$ (Body Frame)
1	Left Δy	Left EE Local Frame 기준 Y축 변화량	$P_{next} - P_{curr}$ (Body Frame)
2	Left Δz	Left EE Local Frame 기준 Z축 변화량	$P_{next} - P_{curr}$ (Body Frame)
3	Left $\Delta Roll$	Left EE 기준 Roll 회전 변화량	<code>as_euler('xyz')[0]</code> (Relative Rot)

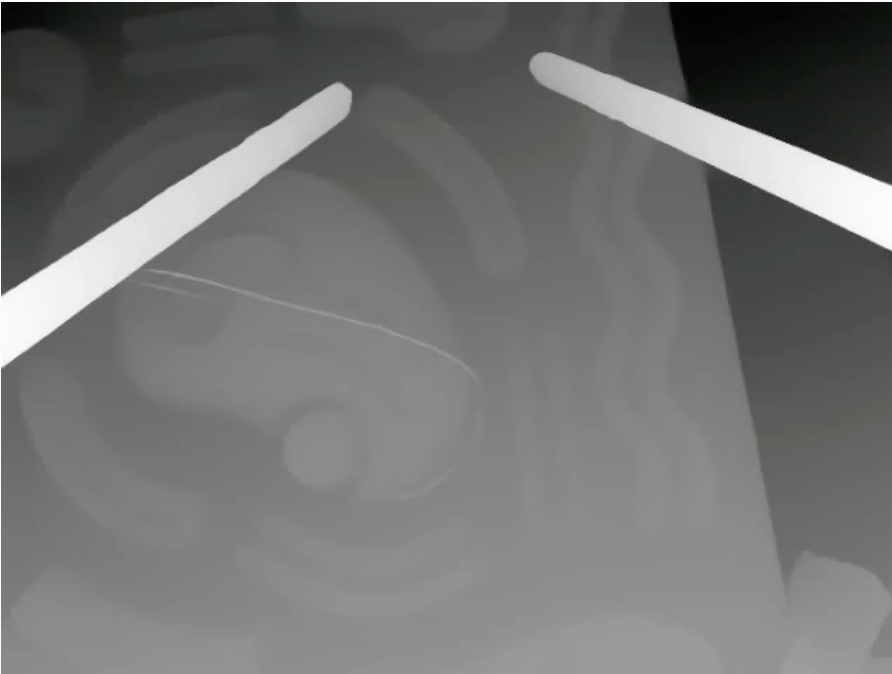
Using 4 variables for position representation
(only 4 needed because of RCM constraint)



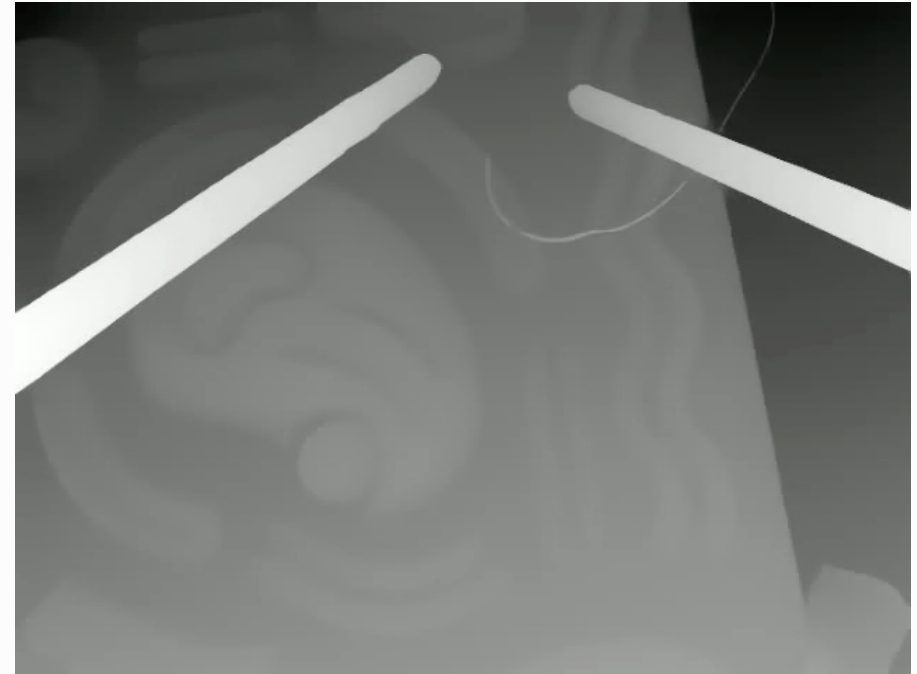
All values are dynamic - Good

Speed-based Filtering

Delta pose can be thought of as speed.
Cannot have lots of stationary times.



Original data – Over 3 minutes long



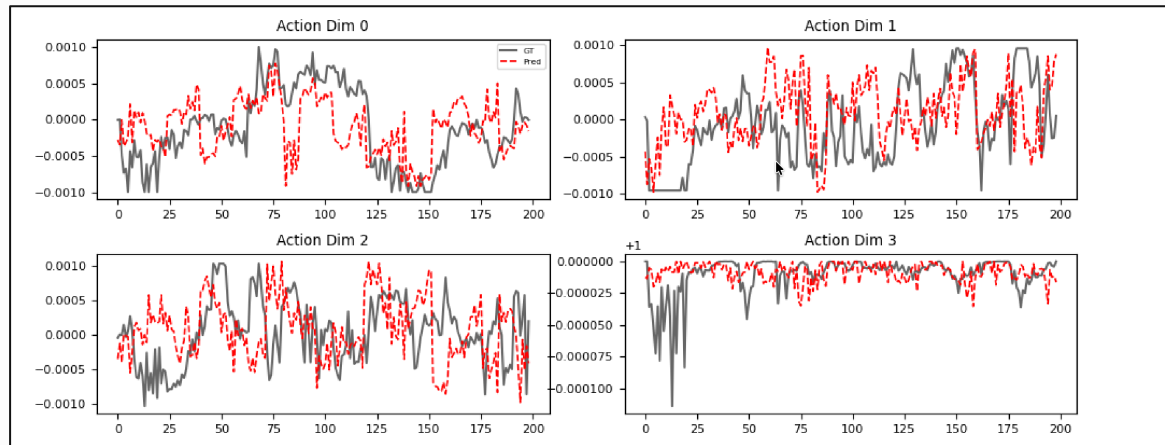
Speed-based filtered data – Less than 60 seconds

Apply Speed filtering with threshold **0.7mm/s, 0.01 rad/s**

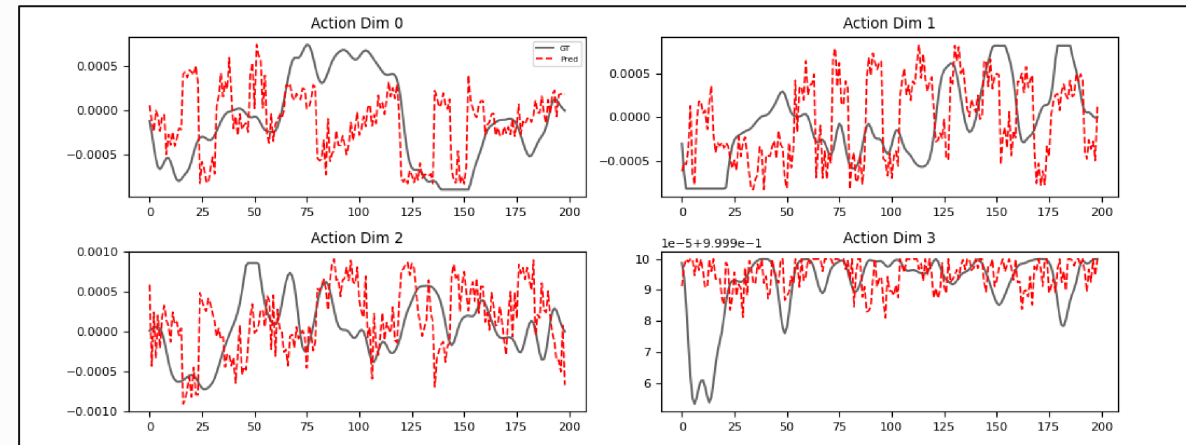
Action Smoothing

Original GT Action graph has lots of sharp points

Low-pass filter to extract important movements, not vibrations



Before smoothing



After smoothing

Apply Gaussian filtering with $\sigma=2.0$ to make smoother GT action